

ARVIAT, NU, Canada

WMO: 711740

Lat: 61.098N

Lon: 94.069W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.4	-37.8	-43.9	0.1	-39.3	-42.3	0.1	-37.8	18.7	-27.4	16.5	-26.0	5.3	330	1.072

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	20.3	14.8	17.7	13.4	15.7	12.4	15.6	19.6	13.9	16.9	12.6	15.3	6.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.8	17.1	12.3	8.9	15.4	11.0	8.2	13.7	43.7	19.6	39.2	17.0	35.8	15.6	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	14.0	12.6	DB	-40.7	25.9	2.4	2.4	-42.4	27.6	-43.8	28.9	-45.2	30.3	-47.0	32.0	(4)
(5)			WB	-40.8	18.6	2.4	1.2	-42.5	19.4	-43.9	20.1	-45.2	20.8	-47.0	21.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.6	-27.3	-28.0	-22.5	-13.9	-4.1	4.7	11.2	10.5	5.5	-2.4	-14.8	-22.9	(6)	
	DBStd	15.16	6.70	6.29	7.61	6.85	4.90	3.93	3.42	2.58	3.40	4.64	7.33	7.25	(7)	
	HDD10.0	6887	1157	1063	1006	717	438	166	26	24	140	384	745	1020	(8)	
	HDD18.3	9816	1415	1297	1264	967	697	409	223	242	385	643	995	1278	(9)	
	CDD10.0	114	0	0	0	0	0	7	62	40	5	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	18	0	0	0	0	0	4	13	1	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	1	1	0	0	0	0	0	(13)	
	WSAvg	6.4	6.9	6.9	6.6	6.4	6.1	5.6	5.5	5.9	6.4	7.0	7.0	6.8	(14)	
(15) Precipitation	PrecAvg	319	14	13	26	18	18	27	40	55	47	31	20	16	(15)	
	PrecMax	503	49	76	156	61	48	55	74	104	128	95	97	52	(16)	
	PrecMin	167	0	0	1	1	1	1	5	10	9	7	1	2	(17)	
	PrecStd	92	14	16	36	15	12	13	19	26	25	21	20	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-9.3	-1.8	1.7	8.8	22.3	25.5	22.2	17.9	8.7	0.7	-3.4	(19)	
		MCWB	-8.0	-9.7	-2.4	1.0	6.5	14.0	16.6	16.4	12.6	7.0	0.4	-3.9	(20)	
	2%	DB	-11.1	-12.7	-5.1	0.1	4.6	16.2	21.1	18.5	13.8	5.4	-0.5	-5.5	(21)	
		MCWB	-11.4	-13.0	-5.5	-0.4	3.2	11.8	15.3	14.1	10.7	4.6	-0.9	-5.9	(22)	
	5%	DB	-14.2	-15.0	-8.0	-1.8	2.8	12.5	19.0	16.4	11.7	4.1	-1.7	-8.7	(23)	
		MCWB	-14.6	-15.3	-8.4	-2.3	1.8	9.2	14.1	12.8	9.5	3.5	-2.2	-9.0	(24)	
	10%	DB	-16.8	-17.4	-10.4	-3.6	1.4	10.1	16.9	14.7	10.0	3.0	-3.0	-11.4	(25)	
		MCWB	-17.2	-17.8	-10.8	-4.0	0.8	7.5	13.1	11.9	8.6	2.3	-3.5	-11.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.0	-9.4	-2.2	1.2	6.4	15.0	17.9	17.3	13.9	7.3	0.5	-3.7	(27)	
		MCDB	-7.9	-9.1	-1.7	1.7	8.7	21.1	22.5	20.3	17.1	8.6	0.8	-3.4	(28)	
	2%	WB	-11.4	-12.9	-5.4	-0.3	3.3	11.9	16.2	14.9	11.2	4.8	-0.8	-5.8	(29)	
		MCDB	-11.1	-12.7	-5.1	0.0	4.5	16.1	20.7	17.8	12.9	5.3	-0.6	-5.4	(30)	
	5%	WB	-14.6	-15.3	-8.3	-2.2	1.8	9.4	14.6	13.3	9.8	3.5	-2.1	-9.0	(31)	
		MCDB	-14.3	-15.0	-8.0	-1.8	2.6	12.4	17.8	15.6	11.3	4.0	-1.8	-8.8	(32)	
	10%	WB	-17.2	-17.7	-10.8	-4.0	0.9	7.6	13.2	12.2	8.6	2.3	-3.5	-11.7	(33)	
		MCDB	-16.9	-17.3	-10.5	-3.6	1.3	10.0	16.4	14.2	9.8	2.8	-3.0	-11.4	(34)	
(35) Mean Daily Temperature Range	MDBR		7.4	7.6	8.8	9.0	5.7	6.1	7.3	6.0	5.2	4.9	7.8	7.4	(35)	
	5% DB	MCDBR	9.7	10.0	10.3	8.5	6.4	11.7	11.9	9.7	8.7	5.0	7.6	8.7	(36)	
		MCWBR	9.6	10.0	10.3	8.3	5.2	7.5	6.7	5.9	5.9	4.4	7.6	8.7	(37)	
	5% WB	MCDBR	9.7	10.3	10.5	8.6	6.2	11.5	11.3	8.5	7.7	4.7	8.0	8.7	(38)	
		MCWBR	9.6	10.2	10.4	8.5	5.1	7.6	7.0	5.8	5.6	4.2	7.9	8.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.192	0.241	0.266	0.335	0.360	0.345	0.369	0.347	0.319	0.281	0.213	0.155	(40)	
	taud		2.058	2.165	2.187	2.012	2.076	2.380	2.429	2.496	2.526	2.451	2.110	1.943	(41)	
	Ebn at Noon		580	758	862	845	854	882	850	843	808	718	528	457	(42)	
	Edh at Noon		38	68	94	139	142	107	100	87	70	53	35	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.38	1.36	3.15	5.26	5.99	6.01	5.54	4.05	2.41	1.02	0.42	0.19	(44)	
	RadStd		0.03	0.06	0.19	0.31	0.52	0.60	0.39	0.28	0.21	0.12	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page